DO NOW — RECALL

Lesson 089 · The Processor City: Architecture Overview

Name: _____ Class: _____ Date: _____

1. What is the function of a program counter?

2. What does the ALU perform?

3. Name one type of bus.

4. A D flip-flop has D=1 and Q=0 just before the active rising edge. What is Q after the edge?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Holds address of next instruction.

2. Model answer

Arithmetic and logical operations.

3. Model answer

Address, data or control bus.

4. Model answer

Q becomes1, as D is sampled at the active edge.

By the end, you can…

• Inside CPU City Hall, the Control Unit organises operations, the ALU carries out arithmetic and logic, and registers hold immediate working state.

• The Memory Warehouse stores program instructions and data. Address, data and control buses connect processor and memory; they carry different kinds of information.

• Von Neumann architecture stores instructions and data in a shared memory/address space, whereas Harvard uses separate instruction and data memories/pathways.

Vocabulary — know these terms

ALU — Arithmetic Logic Unit.

CU — Control Unit.

register — Small fast CPU state storage.

bus — Shared communication pathway.

NEW LEARNING

Read, annotate and explain

Observe

Inside CPU City Hall, the Control Unit organises operations, the ALU carries out arithmetic and logic, and registers hold immediate working state.

Explain

The Memory Warehouse stores program instructions and data. Address, data and control buses connect processor and memory; they carry different kinds of information.

Evaluate

Von Neumann architecture stores instructions and data in a shared memory/address space, whereas Harvard uses separate instruction and data memories/pathways.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Explain a memory read using the city analogy and exact bus names.

Model response

The address identifies where; data is what is transferred. CU coordinates the process.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Name and describe ALU, Control Unit, and registers in processor. [6]

Question 2

Using correct names, explain address, data and control bus during memory read. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate usefulness and limitation of Processor City analogy for Von Neumann architecture. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The data bus carries the address of a memory location.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.